NEET UG 2027: Master Practical Handbook

Practical Chemistry: Qualitative Salt Analysis & Volumetric Redox Titrations

Prepared, Compiled & Authored by Satya Prakash · Calibrated to Revised NMC Syllabus

1 Systematic Qualitative Analysis: Acidic Radicals (Anions)

Class I: Dilute H_2SO_4 Soluble Anions ($CO_3^{2-}, S^{2-}, SO_3^{2-}, NO_2^-$)

Treated with dilute H_2SO_4 to observe volatile evolved gases:

Anion	Gas Evolved	Preliminary Observation	Confirmatory Test & Governing Equation
CO_3^{2-} (Carbonate)	CO_2	Brisk effervescence; odorless, colorless gas.	Turns lime water milky ($CaCO_3\downarrow$); prolonged bubbling clears milkiness forming soluble bicarbonate: $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$.
S^{2-} (Sulfide)	H_2S	Colorless gas with rotten-egg smell.	Turns lead acetate paper black due to formation of lead sulfide ($PbS\downarrow$); gives purple/violet coloration with **sodium nitroprusside**: $S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}$.
SO_3^{2-} (Sulfite)	SO_2	Colorless gas with suffocating odor of burning sulfur.	Turns acidified $K_2Cr_2O_7$ paper green (Cr^{3+}): $K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow Cr_2(SO_4)_3 + K_2SO_4 + H_2O$.
NO_2^- (Nitrite)	NO_2	Pungent-smelling reddish brown fumes.	Starch-iodide paper turns deep blue: $2NO_2^- + 2I^- + 4H^+ \rightarrow 2NO + I_2 + 2H_2O$; gives brown ring immediately with dilute H_2SO_4 and $FeSO_4$ (no conc. acid needed!).

Class II: Concentrated H_2SO_4 Anions ($Cl^-, Br^-, I^-, NO_3^-, CH_3COO^-$)

Treated with concentrated H_2SO_4 and gently heated:

Anion	Observation	Confirmatory Reactions & Diagnostic Identification
Cl^-	Colorless pungent fumes (HCl); dense white fumes with glass rod dipped in NH_4OH (NH_4Cl).	**Chromyl Chloride Test:** Salt $+K_2Cr_2O_7 +$ conc. $H_2SO_4 \xrightarrow{\Delta}$ deep red vapors of chromyl chloride (CrO_2Cl_2); pass into NaOH $\rightarrow$ yellow solution of $Na_2CrO_4 \xrightarrow{Pb(CH_3COO)_2}$ bright yellow precipitate of lead chromate ($PbCrO_4\downarrow$).
Br^-	Reddish brown vapors (Br_2) intensifying on adding MnO_2.	**Organic Layer Test (Layer Test):** Salt solution +chlorine water + $CHCl_3/CCl_4 \xrightarrow{shake}$ organic layer turns **orange-brown** due to liberated free Br_2 dissolved in non-polar layer ($2Br^- + Cl_2 \rightarrow 2Cl^- + Br_2$).
I^-	Deep violet vapors (I_2) turning starch paper blue.	**Layer Test:** Salt solution +chlorine water + $CHCl_3 \xrightarrow{shake}$ organic layer turns **violet**. $AgNO_3$ test gives bright yellow precipitate of AgI insoluble in concentrated NH_4OH.
NO_3^-	Dense brown fumes of NO_2 intensified by adding copper turnings.	**Brown Ring Test:** Salt solution +freshly prepared $FeSO_4 \xrightarrow{\text{trickle conc. } H_2SO_4}$ Brown Ring at junction: $NO_3^- + 3Fe^{2+} + 4H^+ \rightarrow NO + 3Fe^{3+} + 2H_2O$, followed by: $[Fe(H_2O)_6]^{2+} + NO \rightarrow [Fe(H_2O)_5(NO)]^{2+}$ (**Pentaaquanitrosyliron(I)**).
CH_3COO^-	Characteristic vinegar odor.	**Ester Test:** Salt +ethanol + conc. $H_2SO_4 \xrightarrow{\Delta}$ sweet fruity smell of ethyl acetate ($CH_3COOC_2H_5$). **Ferric Chloride Test:** Neutral $FeCl_3 \rightarrow$ deep blood-red coloration ($[Fe_3(OH)_2(CH_3COO)_6]^+$); boiling yields brown basic ferric acetate ppt.

Class III: Independent Group Anions (SO_4^{2-}, PO_4^{3-})

Do not evolve gases with dilute or concentrated acids; identified by direct precipitation:

Sulfate (SO_4^{2-}): $\text{Salt} + BaCl_2 \xrightarrow{\text{dil. HCl}} BaSO_4\downarrow$ (White precipitate insoluble in conc. acid)

Phosphate (PO_4^{3-}): $\text{Salt} + \text{conc. } HNO_3 + (NH_4)_2MoO_4 \xrightarrow{\Delta} (NH_4)_3[P(Mo_3O_{10})_4]\downarrow$ (**Canary Yellow**)

2 Systematic Qualitative Analysis: Basic Radicals (Cations)

Systematic Cation Group Separation Scheme & Group Reagents

Group	Cations	Group Reagent	Precipitated As	Underlying Solubility Product (K_{sp}) Principle
Group 0	NH_4^+	NaOH (heat)	Gas (NH_3)	Identified by Nessler's reagent: brown ppt of iodide of Millon's base ($H_2N-Hg-O-Hg-I$).
Group I	Pb^{2+}	Dilute HCl	Chlorides ($PbCl_2\downarrow$)	Lowest K_{sp} of chlorides; $PbCl_2$ dissolves in hot water, recrystallizes on cooling.
Group II	$Pb^{2+}, Cu^{2+}, As^{3+}$	H_2S gas + dilute HCl	Sulfides ($PbS, CuS, As_2S_3\downarrow$)	**Common Ion Effect:** H^+ suppresses H_2S dissociation, keeping $[S^{2-}]$ very low; only Group II sulfides precipitate.
Group III	Fe^{3+}, Al^{3+}	$NH_4OH + NH_4Cl$	Hydroxides ($Fe(OH)_3, Al(OH)_3\downarrow$)	**Common Ion Effect:** NH_4^+ suppresses NH_4OH dissociation, keeping $[OH^-]$ low to precipitate only Group III hydroxides.
Group IV	$Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+}$	$H_2S + NH_4OH$	Sulfides ($CoS, NiS, MnS, ZnS\downarrow$)	OH^- consumes $H^+ \Longrightarrow$ increases $[S^{2-}]$, exceeding higher K_{sp} of Group IV sulfides.
Group V	$Ba^{2+}, Sr^{2+}, Ca^{2+}$	$(NH_4)_2CO_3 + NH_4OH + NH_4Cl$	Carbonates ($BaCO_3, SrCO_3, CaCO_3\downarrow$)	Carbonates precipitated in alkaline medium; Mg^{2+} remains in solution due to NH_4Cl.
Group VI	Mg^{2+}	$Na_2HPO_4 + NH_4OH$	$Mg(NH_4)PO_4\downarrow$	White crystalline precipitate formed in alkaline ammoniacal medium.

High-Yield Confirmatory Tests & Diagnostic Colorations

1. Nickel Ion (Ni^{2+}) with Dimethylglyoxime (DMG): Rosy-red precipitate stabilized by intramolecular hydrogen bonding:

$$Ni^{2+} + 2\,DMG + 2\,NH_4OH \longrightarrow [Ni(DMG)_2]\downarrow (\text{Rosy Red}) + 2\,NH_4^+ + 2\,H_2O \tag{1}$$

2. Ferric Ion (Fe^{3+}) Confirmatory Tests:

$$\text{With Potassium Ferrocyanide: } 4Fe^{3+} + 3[Fe(CN)_6]^{4-} \longrightarrow Fe_4[Fe(CN)_6]_{\mathbf{3}}\downarrow (\textbf{Prussian Blue})$$
$$\text{With Potassium Thiocyanate: } Fe^{3+} + SCN^- \longrightarrow [Fe(SCN)]^{\mathbf{2+}} (\textbf{Blood Red coloration})$$

3. Copper Ion (Cu^{2+}) Confirmatory Tests:

$$\text{With Excess } NH_4OH: \quad Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{\mathbf{2+}} \quad (\textbf{Azure Blue})$$
$$\text{With Potassium Ferrocyanide: } 2Cu^{2+} + [Fe(CN)_6]^{4-} \longrightarrow Cu_2[Fe(CN)_6]\downarrow \quad (\textbf{Chocolate Brown})$$

4. Flame Test Colors of Alkaline Earth Metals:

$$Ba^{2+} \Longrightarrow \text{Apple Green} \qquad Sr^{2+} \Longrightarrow \text{Crimson Red} \qquad Ca^{2+} \Longrightarrow \text{Brick Red} \tag{2}$$

3 Volumetric Quantitative Redox Titrations

$KMnO_4$ vs Mohr's Salt Titration (Standardization)

Mohr's salt is a primary standard double salt: $(NH_4)_{\mathbf{2}}SO_4 \cdot FeSO_4 \cdot \mathbf{6}H_2O$ (M.W. $= 392\,g/mol$).

$$\text{Oxidation Half: } 5[Fe^{2+} \longrightarrow Fe^{3+} + e^-]$$
$$\text{Reduction Half: } MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$
$$\textbf{Overall Net Redox: } MnO_4^- + \mathbf{5}Fe^{2+} + \mathbf{8}H^+ \longrightarrow Mn^{2+} + \mathbf{5}Fe^{3+} + \mathbf{4}H_2O$$

$$\frac{\mathbf{M_1V_1}}{\mathbf{1}}(KMnO_4) = \frac{\mathbf{M_2V_2}}{\mathbf{5}}(\text{Mohr's Salt}) \Longrightarrow \mathbf{N_1V_1 = N_2V_2} \tag{3}$$

(Valence factor: $n = 5$ for $KMnO_4$, $n = 1$ for Fe^{2+})

Crucial Practical Precautions:

- **Acidifying Agent:** Strictly use **dilute** H_2SO_4.
  - **Why NOT HCl?** Permanganate oxidizes HCl to chlorine gas ($2MnO_4^- + 16HCl \rightarrow 2Mn^{2+} + 5Cl_2\uparrow + 8H_2O + 6Cl^-$), artificially consuming excess $KMnO_4$!

  - **Why NOT HNO_3?** Nitric acid is a strong oxidizing agent itself and competes with $KMnO_4$ in oxidizing Fe^{2+}.
- **Indicator:** $KMnO_4$ acts as a **self-indicator**. The endpoint is marked by the appearance of a **permanent faint pink color** (excess single drop of MnO_4^-).

$KMnO_4$ vs Oxalic Acid Titration

Oxalic acid is a primary standard dihydrate: $(COOH)_{\mathbf{2}} \cdot \mathbf{2}H_2O$ (M.W. = 126 g/mol, Equivalent Weight = 63 g/eq).

$$\text{Oxidation Half: } 5[C_2O_4^{2-} \longrightarrow 2CO_2 \uparrow + 2e^-]$$

$$\text{Reduction Half: } 2[MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O]$$

$$\textbf{Overall Net Redox: } \mathbf{2}MnO_4^- + \mathbf{5}C_2O_4^{2-} + \mathbf{16}H^+ \longrightarrow \mathbf{2}Mn^{2+} + \mathbf{10}CO_2 \uparrow + \mathbf{8}H_2O$$

$$\frac{\mathbf{M_1V_1}}{\mathbf{2}}(KMnO_4) = \frac{\mathbf{M_2V_2}}{\mathbf{5}}(\text{Oxalic Acid}) \implies \mathbf{N_1V_1 = N_2V_2} \tag{4}$$

(Valence factor: $n = 5$ for $KMnO_4$, $n = 2$ for $C_2O_4^{2-}$)

Crucial Practical Precautions:

- **Pre-Heating Requirement:** Oxalic acid solution must be heated to $60 - 70°C$ before starting titration. At room temperature, the reaction is extremely slow.
- **Autocatalysis:** The initial disappearance of pink color is slow, but once the first few drops react, the reaction becomes rapid because product Mn^{2+} **acts as an autocatalyst**!
- **Do NOT boil ($> 70°C$):** High temperature causes thermal decomposition of oxalic acid into CO_2, CO, and H_2O.